Zifu Wan

https://zifuwan.github.io/ zifuw@andrew.cmu.edu (+1) 412-638-9902

EDUCATION

Carnegie Mellon University, Pittsburgh, USA *08/2023 – 08/2025*

- *M.Sc. in Robotics, Robotics Institute, School of Computer Science* GPA: 4.09/4.0

Dalian University of Technology, Dalian, China *09/2019 – 06/2023*

- *B.Eng. in Artificial Intelligence, School of Artificial Intelligence* GPA: 89.4/100

WORK EXPERIENCE

Research Engineer in General Robotics *09/2025 – Present*

Advisor: Dr. Jonathan Huang; Topic: Teleoperation, VLA, and Harness Engineering *Redmond, WA*

- Led the ARMs team, driving PRDs, OKRs, and product roadmap.
- Integrated teleoperation interfaces including GELLO-UR, JoyLo-Galaxea, VR, and Haply for force-aware teleoperation.
- Trained and deployed VLA models (PI05) with RTC across UR and Galaxea embodiments.
- Developed a Real2Sim2Real harness for VLA evaluation and auto-engineering.

Research Assistant in Advanced Agent Robotics Technology Lab, Carnegie Mellon University *08/2023 – 08/2025*

Advisor: Prof. Katia Sycara, IEEE/AAAI Fellow; Topic: Robust/Distributed Perception *Pittsburgh, PA*

- Introduced a comprehensive benchmark for task-oriented part segmentation and affordance learning.
- Developed a Mamba-based fusion method for efficient multi-modal semantic segmentation.
- Proposed two training-free algorithms to mitigate hallucinations in large vision-language models.

Research Intern in AI Innovation Center, Midea Group *02/2023 – 07/2023*

Advisor: Dr. Ning Liu; Topic: Model Compression, Efficient Neural Network Design *Beijing, China*

- Deployed detection/segmentation/3D reconstruction/Automatic Speech Recognition TVM models on internal platform.
- Designed a light-weight semantic segmentation algorithm with magnitude-based pruning and multi-level distillation.
- Optimized ONNX graph before deploying on NVIDIA Jetson Nano platform with TensorRT acceleration.
- Won the most accurate with top-5 speed award in *2023 Low power Computer Vision Challenge* ($1000 award).

Research Assistant in ME Department, Huazhong University of Science and Technology *08/2021 – 09/2021*

Advisor: Prof. Yihua Kang; Topic: Signal Processing, Nondestructive Testing (NDT) *Wuhan, China*

- Studied the relationship between each signal feature and lift-off values extracted from Pulsed Eddy Current (PEC) signals.
- Developed a steel hardness classification method by eliminating the lift-off effect with combined signal features.
- Outperformed traditional methods by over 30% in accuracy for hub bearing groove surface hardness classification.

SELECTED PUBLICATIONS (* *indicates equal contribution*)

- **Zifu Wan***, Ce Zhang*, Silong Yong, Martin Q. Ma, Simon Stepputtis, Louis-Philippe Morency, Deva Ramanan, Katia Sycara, Yaqi Xie. ONLY: One-Layer Intervention Sufficiently Mitigates Hallucinations in Large Vision-Language Models. In *International Conference on Computer Vision (ICCV)*, 2025 [PDF][Code].
- Ce Zhang*, **Zifu Wan***, Zhehan Kan, Martin Q. Ma, Simon Stepputtis, Deva Ramanan, Russ Salakhutdinov, Louis-Philippe Morency, Katia Sycara, Yaqi Xie. Self-Correcting Decoding with Generative Feedback for Mitigating Hallucinations in Large Vision-Language Models. In *International Conference on Learning Representations (ICLR)*, 2025 [PDF][Code], also at NeurIPS 2024 Workshop RBFM [PDF].
- Silong Yong, Venkata Nagarjun Pudureddiyur Manivannan, Bernhard Kerbl, **Zifu Wan**, Simon Stepputtis, Katia Sycara, Yaqi Xie. OMG: Opacity Matters in Material Modeling with Gaussian Splatting. In *International Conference on Learning Representations (ICLR)*, 2025 [PDF].
- **Zifu Wan**, Pingping Zhang, Yuhao Wang, Silong Yong, Simon Stepputtis, Katia Sycara, Yaqi Xie. Sigma: Siamese Mamba Network for Multi-Modal Semantic Segmentation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025 *(Oral Presentation)* [PDF][Code][Website].
- Ce Zhang, **Zifu Wan**, Simon Stepputtis, Katia Sycara, Yaqi Xie. Spectral-Aware Global Fusion for RGB-Thermal Semantic Segmentation. In *IEEE International Conference on Image Processing (ICIP)*, 2025 [PDF].
- **Zifu Wan**, Yaqi Xie, Ce Zhang, Zhiqiu Lin, Zihan Wang, Simon Stepputtis, Deva Ramanan, Katia Sycara. InstructPart: Task-Oriented Part Segmentation with Instruction Reasoning. In *Association for Computational Linguistics (ACL)*, 2025 *(Main conference)* [PDF][Website], also at *AAAI Workshop on Public Sector LLMs*, 2024 [PDF].
- Tianyu Yan, **Zifu Wan**, Xinhao Deng, Pingping Zhang, Yang Liu, Huchuan Lu. MAS-SAM: Segment Any Marine Animal with Aggregated Features. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024 [PDF][Code].

- **Zifu Wan***, Tianyu Yan*, Pingping Zhang, Gong Cheng, Huchuan Lu. TransY-Net: Learning Fully Transformer Networks for Change Detection of Remote Sensing Images. In *IEEE Transactions on Geoscience and Remote Sensing*, 2023 [PDF].
- Tianyu Yan, **Zifu Wan**, Pingping Zhang. Fully Transformer Network for Change Detection of Remote Sensing Images. In *Proceedings of the Asian Conference on Computer Vision (ACCV)*, 2022 (With Student Travel Grant) [PDF][Code].

RESEARCH PROJECTS

Mitigating Hallucinations in Large Vision Language Models (LVLM) with Generative Feedback *07/2024 – 10/2024*

- Discovered the potential of generative models in mitigating hallucinations in LVLMs at both the response and token levels.
- Proposed a training-free algorithm for LVLMs that enhances the accuracy of responses by integrating generative feedback.
- Outperformed other methods in five benchmarks, covering wide areas from image captioning to visual question answering.

Sigma: Siamese Mamba Network for Multi-Modal Semantic Segmentation *02/2023 – 04/2024*

- Proposed the first successful application of state space models, specifically Mamba, in multi-modal semantic segmentation.
- Introduced a Mamba-based fusion method and a channel-aware decoder, to extract and integrate information seamlessly.
- Outperformed other methods on four RGB-Thermal and RGB-Depth benchmarks with superior accuracy and efficiency.

InstructPart: Affordance-based Part Segmentation from Language Instruction *09/2023 – 01/2024*

- Presented the largest real-world task-oriented part segmentation benchmark with hand-labeled instructions and masks.
- Evaluated state-of-the-art vision language models on the benchmark and revealed their limitations.
- Developed a baseline method built upon SOTA foundation models, achieving over a 30% improvement in IoU metrics.

Adapting Segment Anything Model to Marine Animal Segmentation *07/2023 – 11/2023*

- Developed an adapter-informed SAM Encoder with a hypermap extraction module for marine animal feature extraction.
- Proposed a progressive prediction decoder to capture a wide range of global cues and local details.
- Consistently outperformed other methods on four marine animal segmentation benchmarks.

Multimodal Renal Tumor CT Image Detection and Segmentation *02/2023 – 08/2023*

- Introduced a large-scale multimodal benchmark (over 20,000 images from four modalities) for renal tumor identification.
- Benchmarked fully-supervised detection methods, achieving an average of 71.3% in $AP_{50\text{-}95}$ metric.
- Developed a weakly-supervised method for CT image segmentation for computer-aided diagnosis usage.

Transformer-Based Diffusion Modeling for Change Detection of Remote Sensing Images *03/2023 – 07/2023*

- Developed a Swin-Transformer-based diffusion probabilistic model for remote sensing image feature extraction.
- Designed a spatial-aware self-distillation method for multi-level feature enhancement.
- Surpassed other methods on three change detection benchmarks, achieving superior remote sensing image generation.

Fully Transformer Network for Change Detection of Remote Sensing Images *04/2022 – 04/2023*

- Proposed a learning framework for global feature extraction and multi-level feature combination in a pyramid manner.
- Introduced a pyramid structure grafted with a progressive attention module to further improve the feature representation.
- Outperformed most SOTA methods on four public change detection benchmarks with superior efficiency.

Siamese Attentive Convolutional Network for Effective Remote Sensing Image Change Detection *06/2022 – 09/2022*

- Proposed to jointly utilize channel-wise and spatial-wise attention mechanisms to aggregate multi-level features.
- Introduced a method to select more discriminating features in a feature difference view during the decoding phase.
- Outperformed SOTA methods on three change detection benchmarks across five metrics.

Semantic Map Construction for Outdoor Mobile Robots *08/2022 – 02/2023*

- Implemented Simultaneous Localization and Mapping (SLAM) algorithms on the robotic operating system.
- Designed a derivative algorithm of LeGO-LOAM to enhance the feature-matching phase with extra semantic information.
- Deployed the algorithm to a Robomaster AI robot and reached better performance in the mapping process.

Hand Gesture Controlled Robot Car Based on NVIDIA Jetbot platform *11/2021 – 12/2021*

- Embedded a gesture recognition algorithm (based on ResNet) in the NVIDIA Jetson Nano platform.
- Introduced a hand gesture analysis method with real-time image thresholding and key-point extraction.
- Deployed the designed algorithm on an NVIDIA JetBot car, achieving 100% success in gesture control.

PERSONAL SKILLS

- **Computer Skills:** Python, C++, ROS, Linux, LaTeX, Git, Overleaf, Markdown
- **Professional Skills:** Pytorch, OpenCV, ONNX, PyTorch Lightning, TensorRT, Hugging Face

ACADEMIC SERVICES

- Journal Reviewer: *TPAMI*, *IJCV*, *TCSVT*
- Conference Reviewer: *NeurIPS 2025*, *IROS 2025*, *BMVC 2025*, *ICASSP 2025*, *WACV 2025*, *MM 2024*, *ICME 2024*